

Biomarker Discovery by Semantic Technologies

Biomarker Summit 2013

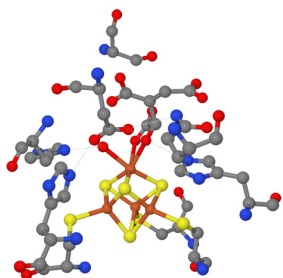
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Progress of e-Discovery in Drug Research



Traditional bioinformatics: structured data

Sequence alignment, gene finding, genome assembly, protein structure prediction, gene expression...



New opportunities: e-Discovery in unstructured data

Knowledge represented as PubMed with 22 million citations, growing at the rate of 1.7 papers/minute

Merck's
question

Is it possible to drive drug research by text mining large pools of biomedical documents?

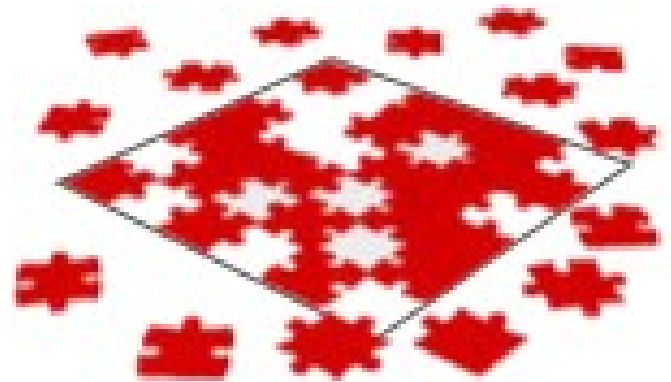
Semantic Technologies in the Pharma Industry

Commonly used: **NLP to extract triples** “*entity 1-relation-entity 2*”
sentence-by-sentence

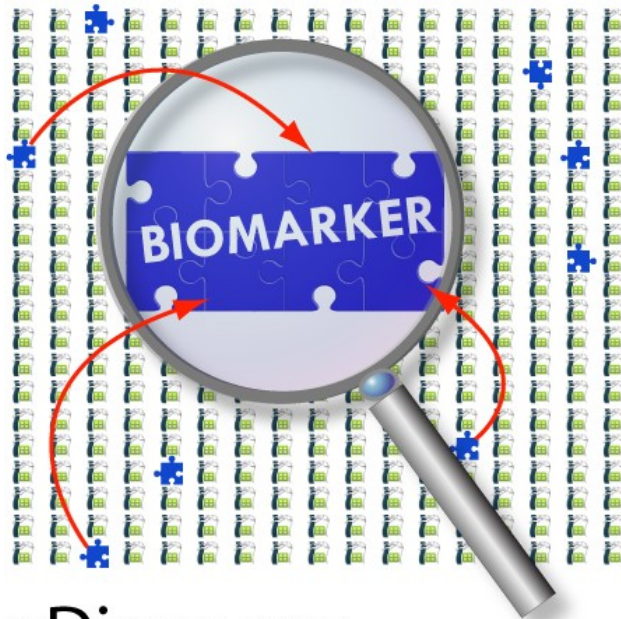
- ⇒ helps to care for ontologies / libraries
- ⇒ finds only what has been written down by an author, i.e.
is not a discovery approach

Going beyond triples

Analyze text collections globally to identify small, seemingly unrelated and unnoticed facts dispersed over isolated texts, like assembling the scattered pieces of a puzzle.



The Experiment of Merck & Co with InfoCodex



eDiscovery

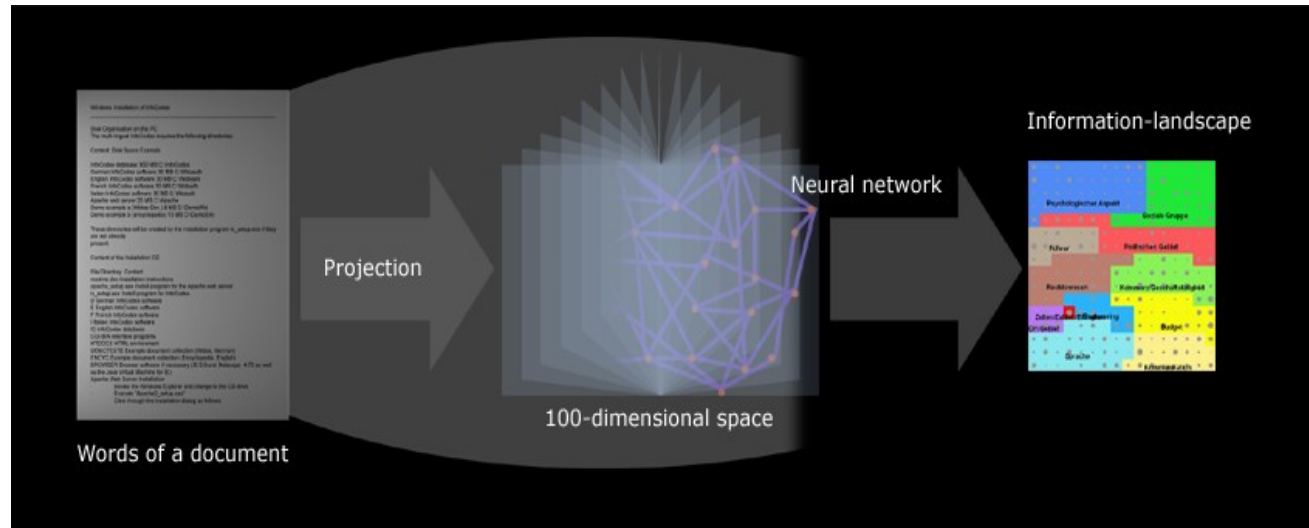
The objective:

- ▶ Test pure machine intelligence for “semantic” drug research

The tasks:

- ▶ Discover novel biomarkers for diabetes and obesity (D&O) by analyzing 120'000 medical publications (PubMed etc.)
- ▶ Blind experiment, no human feedback

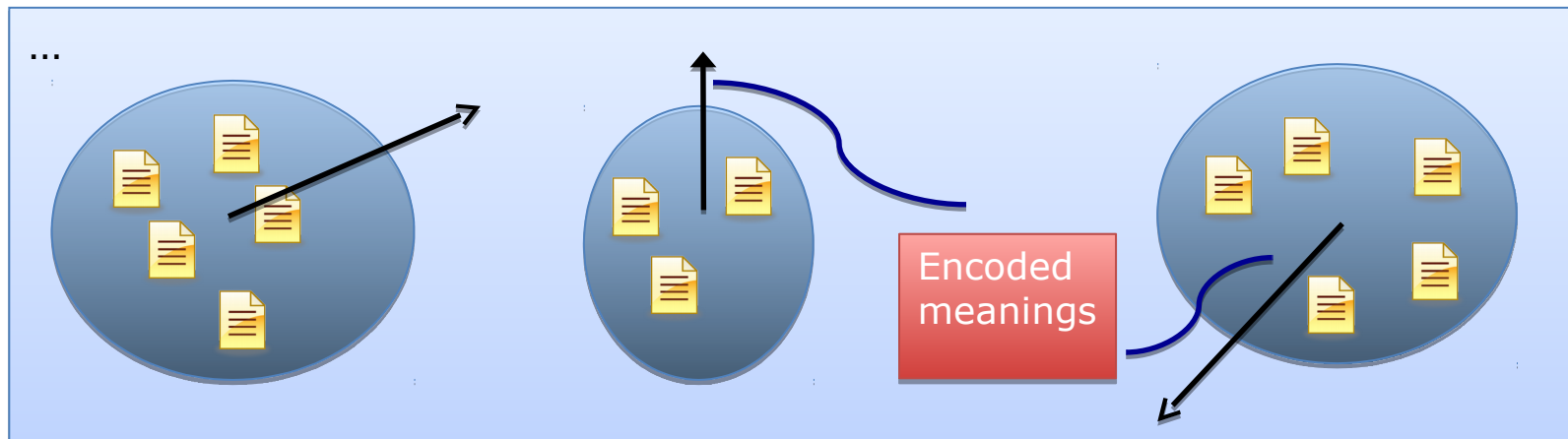
Biomarker: \$ 13.6 billion market in 2011, growing to \$ 25 billion by 2016



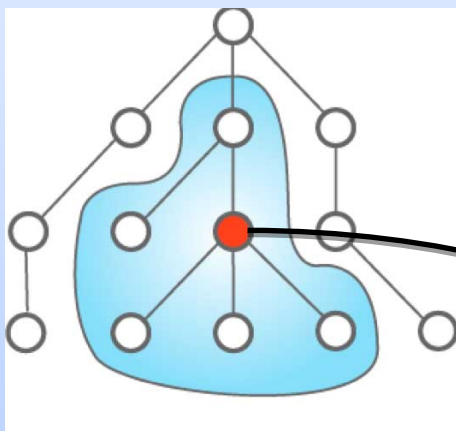
- **Ability to categorize unstructured information**
(in a benchmark, InfoCodex reached the very high clustering accuracy of 88%)
- **Advanced statistics: combination of unnoticed correlations**
(the sentence-by-sentence analysis of the NLP approaches can detect only those relations that have been written down by an author, i.e. that are already known)

Step 1: Establish Reference Models for Biomarkers

- Collect documents describing known biomarkers for diabetes
- Cluster these documents (build groups of similar documents)
- Each cluster is considered as a reference model for the meanings of “biomarkers for diabetes”



Step 2: Determine the Meaning of Unknown Words



Co-occurrences with words in internal knowledge base
 → most probable hypernym → "is a" , "has to do"

A	B	C
Unknown term	Constructed hypernym	Associated descriptor 1
Nn1250	clinical study	insulin glargine
Tolterodine	cavity	overactive bladder
Ranibizumab	drug	macular edema
Nn5401	clinical study	insulin aspart
Duloxetine	antidepressant	personal physician
Endocannabinoid	receptor	enzyme
Becaplermin	pathology	ulcer
Candesartan	cardiovascular disease	high blood pressure
Srt2104	medicine	placebo
Olmesartan	cardiovascular medicine	amlodipine
Hctz	diuretic drug	hydrochlorothiazide
Eslicarbazepine	anti nervous	Zebinix
Zonisamide	anti nervous	Topiramate Capsules
Mk0431	antidiabetic	sitagliptin
Ziprasidone	tranquilizer	major tranquilizer
Psicofarmacologia	motivation	incentive
Medoxomil	cardiovascular medicine	amlodipine

Example:

"Hctz" is a "diuretic drug" and is a synonym of "hydrochlorothiazide"

(estimated by machine intelligence plus the internal knowledge base)

Step 3: Construct Potential D&O Biomarkers (substances close to one of the reference models)

Links to the relevant PubMed documents

	A	B	C	D	E	F	
1	Part "Biomarkers" from Pubmed with confidence level > 5%; 100% refers to biomarkers of the reference set						
2							
3	Term	Relationship	Object	Target	Conf	N.Do	PMIDs
4							
5	Human equilibrative nucleoside transporter-3	BiomarkerFor	Diabetes		100.0	2	20595384, 20032083
6	Human equilibrative nucleoside transporter-3	SynonymOf	hENT3				
7	microRNA	BiomarkerFor	Diabetes		100.0	44	20857148, 21118127, 21335216, 20015039, 20358579, 20364159, 21261648
8	microRNA	BiomarkerFor	Diabetes	FABP_4_aP2	100.0	1	20486779
9	microRNA	BiomarkerFor	Obesity		26.1	58	21355787, 19650761, 21152117, 21118127, 21118894, 20886002, 19188425
10	microRNA	BiomarkerFor	Obesity	FABP_4_aP2	26.1	4	19460359, 18809385, 21291493, 20486779
11	microRNA	BiomarkerFor	Obesity	GPR74	26.1	1	21036322
12	microRNA	BiomarkerFor	Obesity	AMPK	26.1	1	16459310
13	microRNA	SynonymOf	micro-RNA				
14	microRNA	SynonymOf	micro ribonucleic acid				
15	microRNA	SynonymOf	miRNA				
16	microRNA	SynonymOf	miRNA based				
17	microRNA	SynonymOf	MIR126 gene				
18	microRNA	SynonymOf	MiR-126				
19	potassium inwardly-rectifying	BiomarkerFor	Diabetes		100.0	50	20042013, 20194712, 20368737, 20401705, 20531501, 20546293, 20863361
20	potassium inwardly-rectifying	BiomarkerFor	Diabetes	FTO	100.0	8	18597214, 19020324, 18984664, 20503258, 18598350, 20142250, 18710364
21	potassium inwardly-rectifying	BiomarkerFor	Obesity		21.0	24	20049090, 20307313, 18598350, 18710364, 20712903, 18498634, 21391351
22	potassium inwardly-rectifying	BiomarkerFor	Obesity	FTO	21.0	4	20049090, 18598350, 18710364, 20929593
23	potassium inwardly-rectifying	SynonymOf	KCNJ11				
24	potassium inwardly-rectifying	SynonymOf	Kir6.2 gene				

Assessment of the Results

See Trugenberger et al. BMC Bioinformatics 2013, 14:51

<u>Term</u>	<u>Relat.</u>	<u>Object</u>	<u>Target</u>	<u>Conf%</u>	<u>#Docs</u>
wenqing	BiomarkerFor	Obesity	Obesity	53.5	29
proteomic	BiomarkerFor	Obesity	Obesity	40.8	128
gene expression	BiomarkerFor	Obesity	Obesity	38.9	62
Mouse model	BiomarkerFor	Obesity	Obesity	19.8	17
muise	BiomarkerFor	Obesity	Obesity	17.5	20
athero-	BiomarkerFor	Obesity	Obesity	16.5	6
shrna	BiomarkerFor	Obesity	Obesity	9.6	4
inflammation	BiomarkerFor	Obesity	Obesity	8.2	4
TBD	BiomarkerFor	Obesity	Obesity	7.4	3
body weight	PhenoTypeOf	Diabetes	MGAT2		1
cell line	BiomarkerFor	Diabetes	MGAT2		1

Weak Points

Many uninteresting candidates

⇒ too much noise
(can be easily eliminated)

Strong Points

Lots of ***"needles in the haystack"***
 Tens of extremely interesting and valuable candidates

<u>Term</u>	<u>Relat.</u>	<u>Object</u>	<u>Target</u>	<u>Conf%</u>	<u>#Docs</u>
PhenoTypeOf	Obesity	Obesity		7.7	4
PhenoTypeOf	Obesity	Obesity		7	6
BiomarkerFor	Obesity	Obesity		4.9	1
BiomarkerFor	Obesity	Obesity		4.9	1
BiomarkerFor	Obesity	Obesity		2.9	2
BiomarkerFor	Obesity	Obesity		2.2	1
BiomarkerFor	Obesity	Obesity		2.2	1
BiomarkerFor	Obesity	Obesity		2.2	1
BiomarkerFor	Diabetes	Diabetes		14.5	1
BiomarkerFor	Diabetes	Diabetes		2.8	2

Novel and semantically coherent terms, and therefore potentially valuable

(Merck proprietary terms hidden)

Conclusions from the Experiment

Discovery of novel biomarkers

Find potential compounds in drug research

Early recognition of side effects

Competitive advantages / Cutting costs